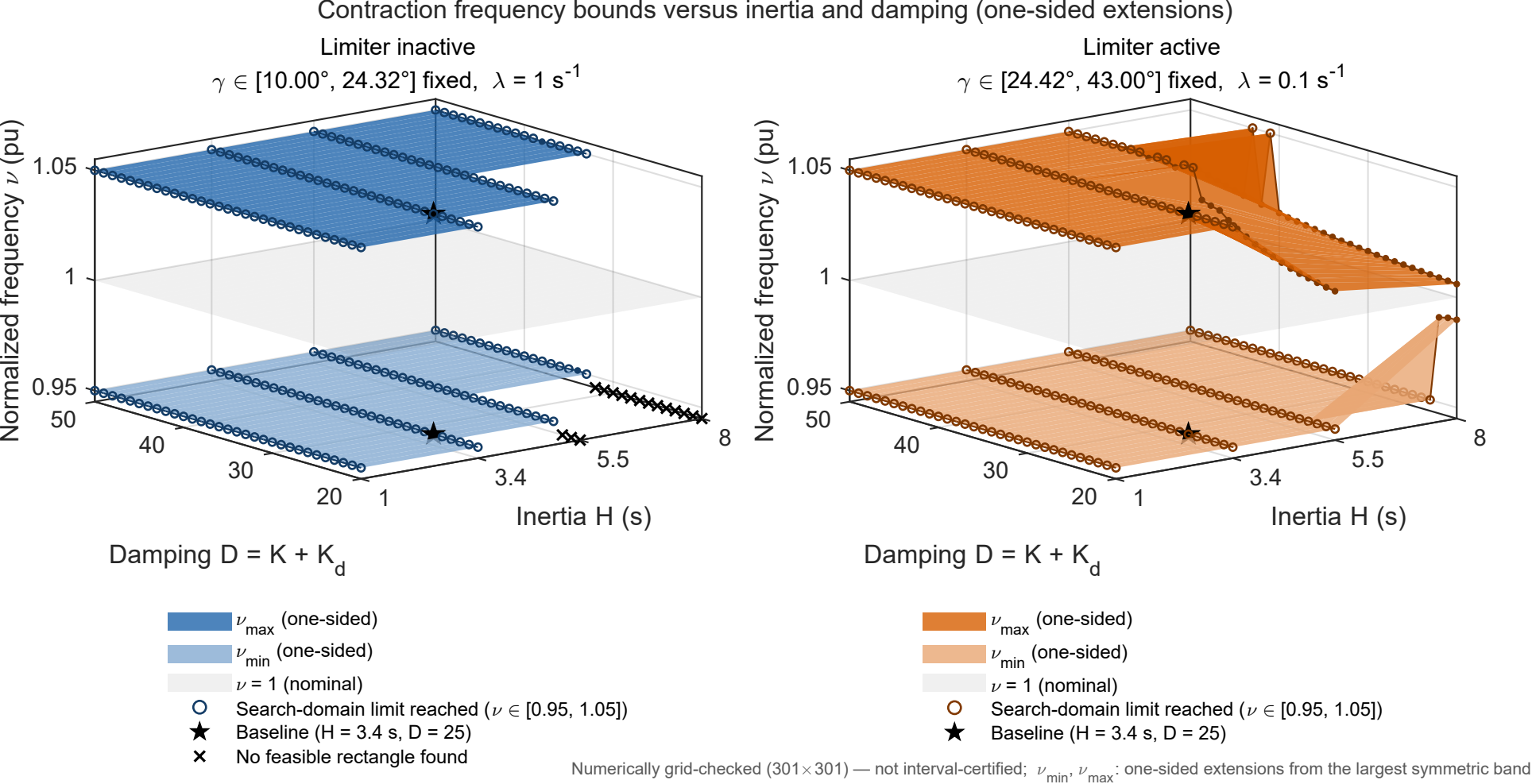

Contraction frequency bounds versus inertia and damping (one-sided extensions)
Limiter inactive
γ ∈ [10.00°, 24.32°] fixed, λ = 1 s-1
Limiter active
γ ∈ [24.42°, 43.00°] fixed, λ = 0.1 s-1
Normalized frequency ν (pu)
1.05
1
0.95
50
40
30
20
Damping D = K + Kd
1
3.4
5.5
8
Inertia H (s)
νmax (one-sided)
νmin (one-sided)
ν = 1 (nominal)
Search-domain limit reached (ν ∈ [0.95, 1.05])
Baseline (H = 3.4 s, D = 25)
No feasible rectangle found
Numerically grid-checked (301×301) — not interval-certified; νmin, νmax: one-sided extensions from the largest symmetric band